

Suman Jash

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SUMMARY

Computer Science student (Java/Python/C++, DSA) who has shipped full-stack applications end-to-end – REST APIs, relational schema design, automated testing, and CI/CD deployment – with experience integrating third-party APIs into production systems.

EDUCATION

Narula Institute of Technology

B.Tech in Computer Science and Engineering, CGPA: 8.99/10

2023 – 2027

Kolkata, West Bengal

Arambagh Vivekananda Academy

CBSE Class XII – 85.8% | CBSE Class X – 94.4%

2021 – 2023

Arambagh, Hooghly

TECHNICAL SKILLS

Languages: Java, Python, C++, C, SQL, JavaScript (ES6+), TypeScript, PHP, HTML5, CSS3

Core CS Concepts: Data Structures & Algorithms, Object-Oriented Programming, System Design, DBMS

Backend & APIs: RESTful API Design, FastAPI, Flask, Node.js, Express.js, Next.js

Databases: PostgreSQL, MySQL, SQLite, MariaDB, Relational Schema Design

DevOps & Testing: Git, GitHub Actions (CI/CD), Docker, Docker Compose, pytest, Playwright, Linux

Security: Authentication, Role-Based Access Control (RBAC), Rate Limiting, Password Hashing, API Key Hashing/Encryption

PROJECTS

- **ShetBhav – Market Intelligence & ERP Platform** | *Python, FastAPI, Next.js, TypeScript, PostgreSQL, Docker, GitHub Actions*
[GitHub](#) | [Live Demo](#) | [API Docs](#)
 - Built a full-stack ERP platform with a modular REST backend (104 endpoints) and a normalized 45-table PostgreSQL schema, serving 4 distinct roles under server-side RBAC.
 - Designed a multi-mode fallback adapter for government market data ingestion (data.gov.in AGMARKNET, 800+ records), keeping the pipeline available through upstream outages and format changes.
 - Automated core business logic – price forecasting, an 8-factor decision engine, and a multi-party negotiation flow – to remove manual work from order and offer tracking.
 - Maintained code quality with 246 backend pytest units and 15 Playwright E2E tests wired into a GitHub Actions CI/CD pipeline for automated build, test, and deploy.
- **CourierAI – Automated Delivery Prediction & Service Platform** | *Python, Flask, scikit-learn, SQLite, Docker, GitHub Actions*
[GitHub](#) | [Live Demo](#)
 - Built a secure prediction REST API (POST /api/predict) with per-user API keys (SHA-256 hashed at rest, Fernet-encrypted for owner reveal), tiered rate limiting, and quota enforcement for high stability under load.
 - Tuned and deployed a HistGradientBoostingRegressor on 49,639 real delivery records via randomized search with 5-fold CV, achieving a mean absolute error of 0.536 days on holdout.
 - Containerized the service with Docker and built a GitHub Actions CI pipeline with health and metrics endpoints for continuous monitoring and deployment.
- **AeroBook – Airline Operations & Reservation System** | *PHP 8, MySQL, JavaScript, Bootstrap 5, Docker*
[GitHub](#) | [Live Demo](#)
 - Integrated live flight and airport data from the AviationStack third-party API into an internal booking system, with automated sync logging and diagnostic checks to catch integration failures early.
 - Designed a normalized 24-table MySQL schema with FK constraints and indexing to support a multi-role platform, including a dedicated Admin Operations Center for internal staff.
 - Hardened the application with prepared statements against SQL injection, CSRF validation, bcrypt password hashing, and session management.

ACHIEVEMENTS & CERTIFICATIONS

- **Competitive Programming:** Codeforces Pupil (1330) | CodeChef 3-Star (1628) | 600+ algorithmic problems solved across LeetCode, CodeChef, and Codeforces.
- **Hackathon Leadership:** Led team in internal Smart India Hackathon round to design a College ERP System; owned system architecture, automated test design, and the technical presentation.
- **Competitive Coding Rank:** Secured 2nd rank in *Code It*, a duo-format coding contest (40 teams in finals).
- **Certifications:** NPTEL Programming in Java (Elite, 94/100) | NPTEL Cloud Computing (92/100) | EduSkills AICTE Virtual Internships – AI Deployment & Automation (10 weeks), Prompt Engineering for AI (8 weeks), DevOps & Cloud Automation (8 weeks).